

Cloud Email Analyzer

Event-driven email threat triage with ML-assisted classification and attachment scanning

TODO: Author/team

TODO: Scientific coordinator

TODO: University / faculty / group



Email evidence is fragmented before review.



Raw messages are messy

MIME variants, envelope recipients, URLs, and attachments all carry evidence.



Signals arrive separately

Classifier output, antivirus results, and mail verdicts keep their source.



Review needs accountability

Scoped access and audit records are required for sensitive review actions.

Normalize once, then analyze and review with context.



Canonical MailReceived event carries metadata and storage pointers, not raw email bodies.

Ingress, analysis, persistence, and review are separated.

AWS mail ingress and event boundary



Analysis, persistence, and review



The event carries metadata and pointers; raw bodies and attachment bytes stay in S3.

Each stage leaves evidence for the next decision.



Recipient ownership

Envelope recipients become user inbox rows or admin review.

Detector outputs

ML verdicts and attachment scan results remain separately visible.

Final state

Aggregation produces the compact verdict shown in the dashboard.

The built system maps responsibilities to small modules.

Cloud services

- Route53 + SES
- S3 object storage
- EventBridge
- Step Functions
- DynamoDB

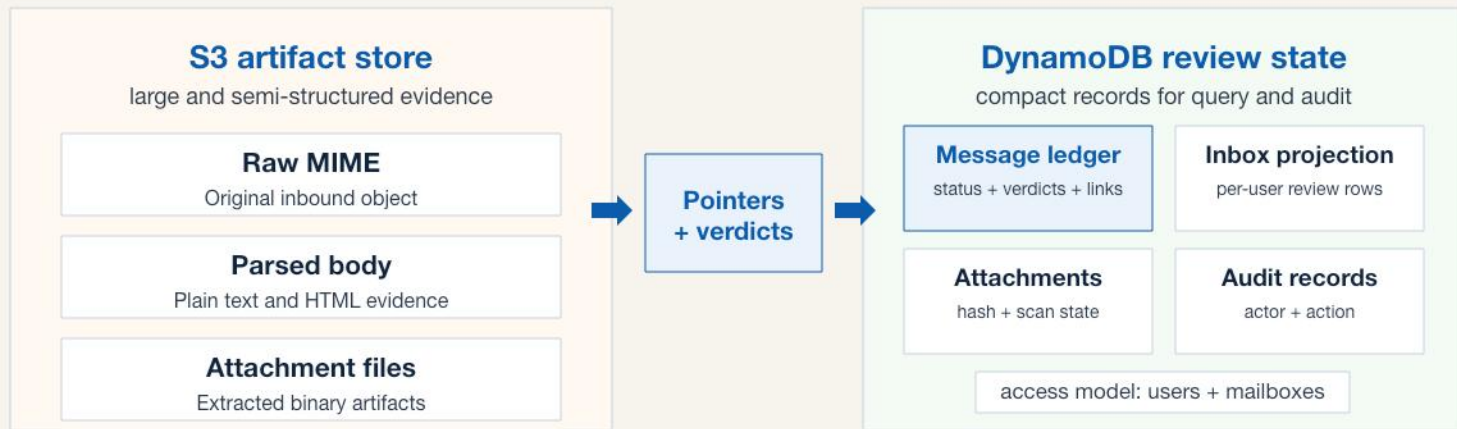
Analysis stages

- InitLedger
- ResolveRecipients
- ParseEmail
- PhishingML
- AttachmentScan
- AggregateVerdicts

Review layer

- FastAPI routes
- HS256 JWT authorization
- React dashboard
- Admin review
- Audit records

Separate artifacts from review state.



The dashboard reads compact state and retrieves artifacts through authorized API calls.

Checked ML artifacts support the packaged classifier.



Evidence scope: training-bundle artifacts, not live organizational traffic.

0.9803

Internal validation accuracy

TN 7748, FP 158, FN 152, TP 7649.

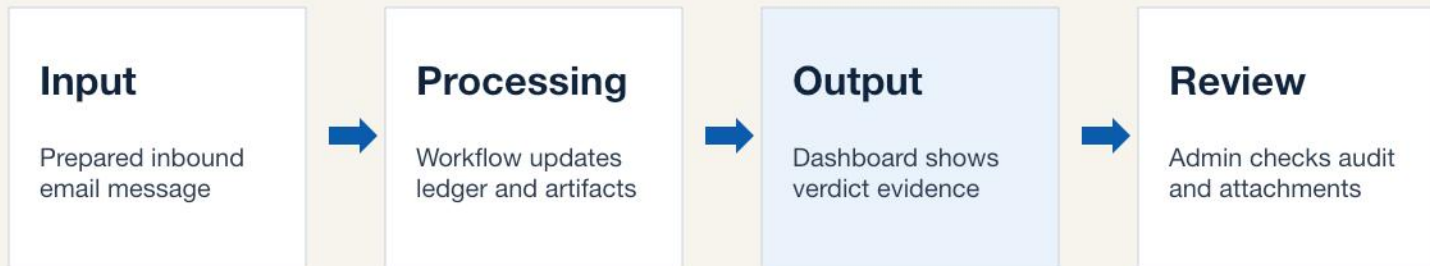
0.9457

Nazario split class-1 recall

FN 17, TP 296.

Takeaway: SGD is the best model among the four plotted classifiers, with accuracy, precision, recall, and F1 near 0.983.

Three minutes: from inbound message to reviewable verdict.



Expected result: one message page with ML verdict, attachment state, hashes, timeline, and audit events.

The architecture leaves room for stronger evidence sources.

Current limits

- Detector scope: text classifier and ClamAV file scan.
- Policy choices: retention, quarantine, escalation, risky HTML handling.
- Lifecycle: managed identity and scanner updates.

Future work

- Add URL/domain reputation and header-auth scoring.
- Move identity management to Cognito.
- Manage ClamAV signature updates.
- Add notes, case assignment, and exportable reports.

Questions

Cloud Email Analyzer turns inbound mail into traceable security evidence.

- AWS-oriented event boundary
- ML classification plus attachment scanning
- Authenticated user/admin review with audit records

TODO: contact / repository link